MIL-E-1/1094B
8 November 1968
SUPERSEDING
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MIL-E-1/517
14 January 1954

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6BJ6A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, RF, remote cutoff

Outline	---	5-2 (EIA) Ⓑ
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k	h	h	a	g2	g3 int sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Pp	Pg2	Alt Ⓑ
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms	W	W	ft
Maximum:	6.9	330	---	135	---	100	---	2.75	0.4	(see note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	250	0	100	0	---	80	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 20 Vdc; Rk = 0 (see note 2)	0.65	II	Is	45	---	mAdc
1256	Electrode current (anode)		0.65	II	Ib	7.2	11.2	mAdc
1268	Total grid current	Ec1 = -1.0 Vdc; Rk = 0 (see note 2)	0.65	II	Ic1	0	-1.0	µAdc
1306	Transconductance (1)		0.65	II	Sm	3,100	4,500	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	6.5	Code I	Ep	---	400	mVac
1256	Electrode current (screen grid)		2.5	I	Ic2	2.3	4.3	mAdc
1301	Heater current		2.5	I	If	138	162	mA
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	Sm	2,750	---	µmhos
1306	Transconductance (3)	Ec1 = -20 Vdc; Rk = 0	6.5	I	Sm	1	50	µmhos
1331	Direct-interelectrode capacitance	No shield	6.5	Code E	Cg1p Cin Cout	--- 3.0 3.5	0.0035 6.0 6.5	pF pF pF
1336	Heater-cathode leakage		2.5	I	Ihk	---	20	µAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
1211	Insulation of electrodes		4.0	S3	R	10	---	Meg

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1501	Intermittent life	Group A; Ehk = 100 V; Rg1 = 0.5 Meg; Ec1 = -1.0 Vdc; Rk = 0	---	---	---	---	---	---	---	---
---	Life-test end point (intermittent) (500 hours)	Transconductance (1)	---	---	---	---	Sm	2,470	---	µmhos

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
1511	Cathode interface life test	Ef = 6.9 V; other electrodes disconnected (see note 3)	---	---	---	---	---	---	---	---
---	Cathode interface life-test end point	Ef = 5.7 V; Eb = Ec2 = Ec3 = 70 Vdc; Ec/Ib = 4.0 mA (see note 4)	---	---	1	3	R	---	70	Ohms

NOTES:

1. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage. Ⓑ
2. This test to be performed at the conclusion of the holding period.
3. An acceptable quality of tubes with respect to the formation of cathode interface resistance shall be established by subjecting 20 tubes to a cutoff life test specified in method 1511. In the event of the rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The life-test sample shall be read at 500 hours (plus 48 hours, minus 24 hours).
4. No other test shall be made from the start of the interface life test until after the measurement of life-test end point (interface).

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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